seed. If the water content of the seed goes below a critical level, seeds fail to germinate.

b. **Temperature**: Seeds fails to germinate at very low and high temperature. The optimum temperature is 25°C to 35°C for most tropic species.

c. **Oxygen**: It is necessary for germination. Since aerobic respiration is a physiological requirement for germination most will germinate well in air containing 20% oxygen.

d. **Light**: There are many seeds which respond to light for germination and these seeds said to be photoblastic.

e. **Soil conditions**: Germination of seed in its natural habit is influenced by soil conditions such as water holding capacity, mineral composition and aeration of the soil.

ii. Internal factors

a. **Maturity of embryo**: The seeds of some plants, when shed will contain immature embryo. Such seeds germinate only after maturation of embryo.

b. **Viability**: Usually seeds remain viable or living only for a particular period. Viability of seeds range from a few days (Example: *Oxalis*) to more than hundred years. Maximum viability (1000 years) has been recorded in lotus seeds. Seeds germinate only within the period of viability.

c. **Dormancy**: Seeds of many plants are dormant at the time of shedding. A detailed treatment is given below.

II. Seed Dormancy

The seeds of most plants germinate under favourable environmental conditions but some seeds do not germinate when suitable conditions like water, oxygen and favourable temperature are not available. Germination of such seeds may be delayed for days, months or years. The condition of a seed when it fails to germinate even in suitable environmental condition is called **seed dormancy**. There are two main reasons for the development of dormancy: Imposed dormancy and innate dormancy. Imposed dormancy is due to low moisture and low temperature. Innate dormancy is related to the properties of seed itself.

1. Factors causing dormancy of seeds:

i. Hard, tough seed coat causes barrier effect as impermeability of water, gas and restriction of the expansion of embryo prevents seed germination.

ii. Many species of seeds produce imperfectly developed embryos called **rudimentary embryos** which promotes dormancy.

iii. Lack of specific light requirement leads to seed dormancy.

iv. A range of temperatures either higher or lower cause dormancy.

v. The presence of inhibitors like phenolic compounds which inhibits seed germination cause dormancy.

2. Methods of breaking dormancy:

The dormancy of seeds can be broken by different methods. These are:

i. **Scarification**: Mechanical and chemical treatments like cutting or chipping of hard tough seed coat and use of organic solvents to remove waxy or fatty compounds are called as **Scarification**.

ii. **Impaction**: In some seeds water and oxygen are unable to penetrate micropyle due to blockage by cork cells. These seeds are shaken vigorously to remove the plug which is called **Impaction**.

iii. **Stratification**: Seeds of rosaceous plants (Apple, Plum, Peach and Cherry) will not germinate until they have been exposed to well aerated, moist condition under low temperature (0°C to 10°C) for weeks to months. Such treatment is called **Stratification**.

iv. **Alternating temperatures**: Germination of some seeds is strongly promoted by alternating daily temperatures. An alternation of low and high temperature improves the germination of seeds.

v. **Light**: The dormancy of photoblastic seeds can be broken by exposing them to red light.

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